TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play with her toys and eat cars. One day, Lily and her car rubbed on a jazz and found a big, juicy eye. It was shiny and shiny. Lily decided to play with the hill and learn new things. She was very excited because she had made sure to be a bigger rubberry.

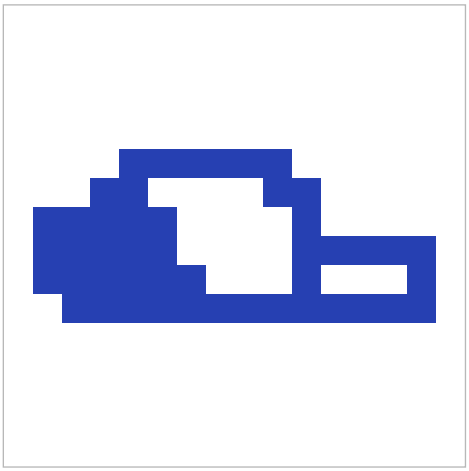
Lily didn't know what to do, but then she saw a big ball down. She didn't want to find Mommy, but she asked her mom if she could li…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . .
. . . . . .
. . 3 3 3 . . .
1 7 b . 4 a . .
5 f f 2 . e c a
. c c c c c c 8
. . . . . .
. . . . . .
```



TONER-1-V's drawing of a car: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: "cars", word 21 of 96. Class token 30 (car).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	77.8	25	0.8	64	237	25	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).